

PORTFOLIO / 2026

Denis Irinnyakov

Data Engineer

Reliable pipelines, streaming systems, analytical data layers, and ML project work.

Moscow / Remote

github.com/karnaksp

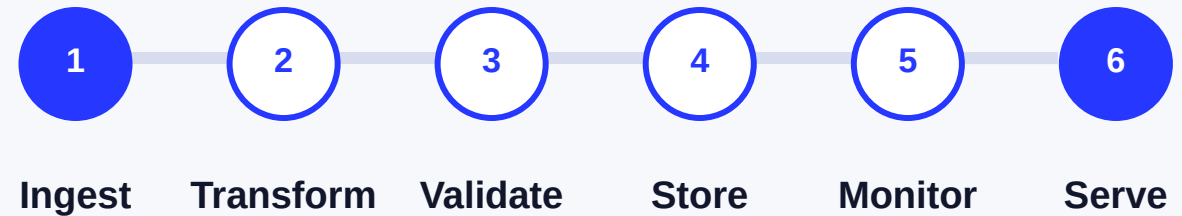


I build data systems across the full lifecycle

From ingestion to product use, with reliability and documentation in view.

I am a data engineer with experience in analytics, backend development, and ML projects. My focus is reliable ETL/ELT, event streaming, analytical DWH/lakehouse layers, data quality, observability, and clear operational documentation.

The projects show more than code: data contracts, run scenarios, checks, architecture, and constraints.



A complete data path, not an isolated task

The toolkit spans data systems and delivery

Grouped from the skills listed in the source portfolio.

Pipelines & orchestration

Airflow
Dagster
dbt
SQLMesh
ETL / ELT

Streaming & processing

Kafka
Redpanda
Spark
Debezium
CDC

Data platforms

PostgreSQL
ClickHouse
Greenplum
BigQuery
MinIO / S3

Engineering & delivery

Python + SQL
Bash + PL/pgSQL
Docker + CI
FastAPI + CLI
Data quality + observability

Four engineering projects anchor the portfolio

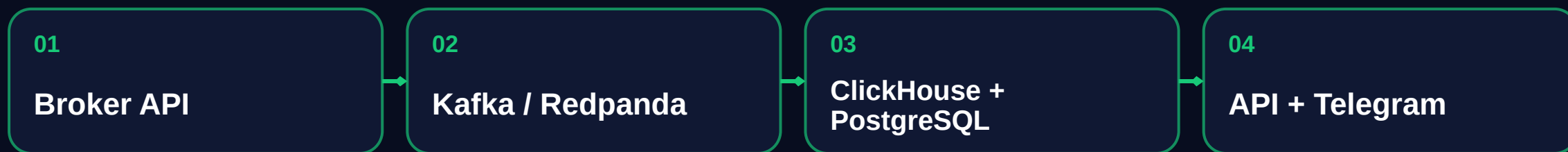
Each covers a distinct point in the path from events to usable data.

01	investment-signals	Real-time market data collection, validation, storage, API, and Telegram delivery.
02	data-forge	Retail CDC pipeline from PostgreSQL through streaming processing to analytical layers.
03	streamify	Music-service event analytics with Kafka, Spark Streaming, Airflow, dbt, and BigQuery.
04	dbt-af	A reproducible and testable path for running dbt models from Airflow.

Six additional projects extend the story into market visualizations, ML hackathons, C/C++, and document automation.

investment-signals observes market data

Collection, validation, history, and delivery in one service.



PURPOSE

Collect and validate quotes and events in real time.

PRACTICAL VALUE

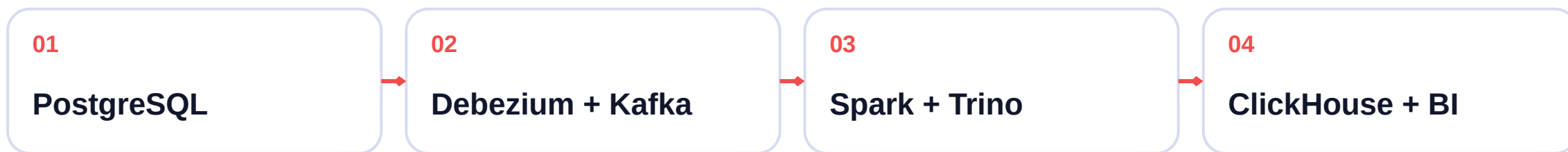
Notice market deviations and ingestion failures sooner while retaining calculation history.

Python / Kafka / Redpanda / ClickHouse / PostgreSQL / FastAPI / Dagster / Grafana / Docker

github.com/karnaksp/investment-signals

data-forge builds a complete retail CDC path

An applied learning environment for a complete CDC pipeline.



PURPOSE

Stream changes from a retail transactional database into processing and analytical layers.

PRACTICAL VALUE

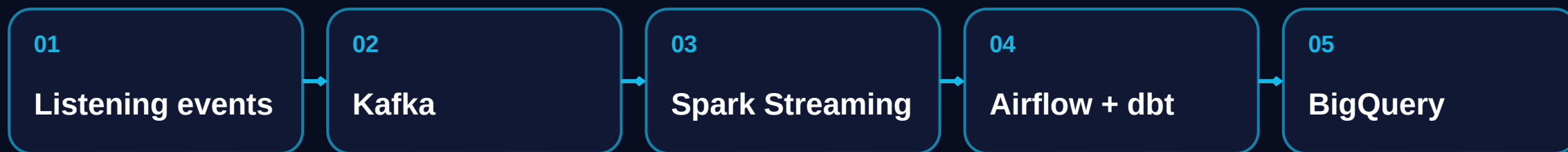
Show the full path from source transactions to validated data marts and reports.

PostgreSQL / Debezium / Kafka / Spark / Trino / MinIO / ClickHouse / Airflow / Superset / Docker Compose

github.com/karnaksp/data-forge

streamify turns listening events into analytics

Streaming, scheduled transformation, and data-mart checks.



PURPOSE

Build an analytics pipeline for music-service listening events.

PRACTICAL VALUE

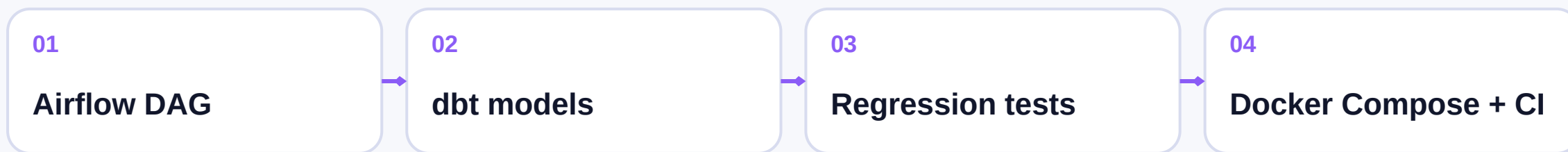
Demonstrate event analytics where streaming, scheduling, and data-mart quality control all matter.

Kafka / Spark Streaming / Airflow / dbt / BigQuery / Docker Compose / GitHub Actions

github.com/karnaksp/streamify

dbt-af makes orchestration reliable

A focused project on repeatable, testable dbt execution from Airflow.



WHAT CHANGED

The manual-run scenario was corrected, checks were added, and the root cause was documented.

PRACTICAL VALUE

Bring orchestration to a state that can be checked, repeated, and maintained.

Python / Airflow / dbt / Docker Compose / GitHub Actions / pytest

github.com/karnaksp/dbt-af

stock-prices automates market visualizations

A CLI and Telegram workflow for ready-to-share animated price reviews.



INTERACTION

Run from the command line or through a Telegram bot.

PRACTICAL VALUE

Prepare a clear price overview without assembling charts manually.

Python / Telegram Bot API / CLI / Docker / MkDocs / GitHub Actions

github.com/karnaksp/stock-prices

Three hackathons show applied ML range

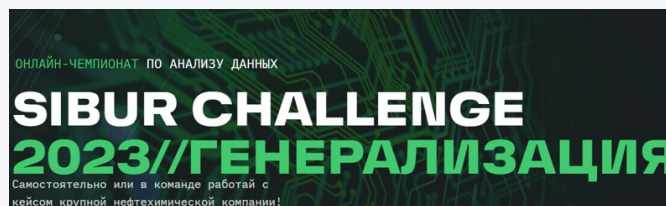
Forecasting tasks across agriculture, industry, and lubricants.



AgroCode Data Science Cup 2023

Animal productivity forecasting using feature preparation, gradient-boosting ensembles, feature selection, and regression evaluation.

CatBoost / XGBoost / LightGBM / stacking / blending



Sibur Challenge 2023

Industrial mixture parameter prediction with data cleaning, LSTM/ML models, blending, and reproducible experiment packaging.

LSTM / deep learning / tabular data / Docker



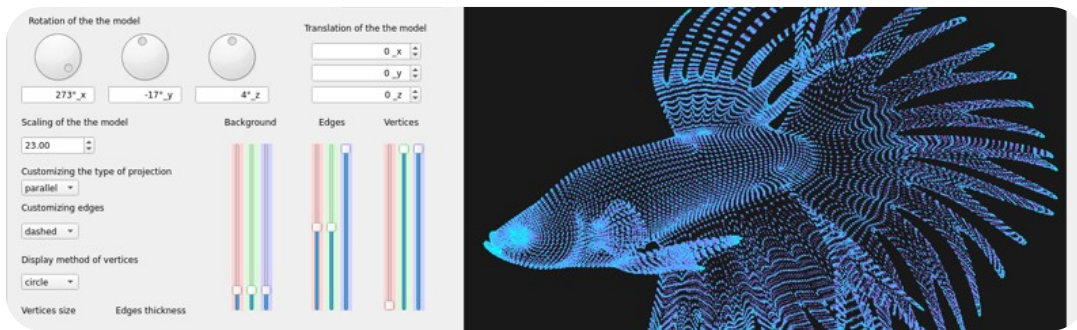
Neftekod 2024

Lubricating-oil parameter prediction for a variable number of components using SMILES and anonymized experimental parameters.

Deep learning / preprocessing / OOP / teamwork

Engineering foundations reach beyond data

Two earlier projects document systems programming and practical automation.



School 21 programming projects

C/C++, structured and object-oriented programming, algorithms, memory, containers, GUI/QT, and team development.



PDF Parser

A small Python automation project for text extraction, regular expressions, procedural processing, and structured output.

Experience spans data and LLM work

Two listed roles connect analytics, warehousing, and LLM-based chatbots.

2023-2024



Data Analyst / DirectService

Loyalty-program analysis; ETL pipeline and PostgreSQL data warehouse setup; migration from ClickHouse and Greenplum; BI reporting visualization.

SQL / Python / Analytics

2024



Data Scientist / Spacecode

Development of chatbots based on large language models.

Machine learning / Data analysis / LLM

Explore the code, architecture, and constraints

Repositories and contact details are available directly from the portfolio.

Education

2023-2024

Programmer, School 21 - project-based programming education from Sber.

2019-2023

PhD, FSC of Biodiversity - fundamental ecology.

2023-2024

Sapiens Solutions - course: Working with Greenplum Database.

Portfolio

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GitHub

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